西南交通大学

土木工程地质课内实习指导书

班　　级 _____

学　　号 _____

姓　　名 _____

指导老师 _____

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1 矿物鉴定实习

1.1 主要鉴定指标及获取方法

（1）光泽：矿物表面反射光线的能力。

通过肉眼观察分辨金属光泽和非金属光泽。

（2）硬度：矿物抵抗外力压入或研磨的能力。

采用小铁刀（硬度为 5.5-6）刻划，若可以被小铁刀刻划出刻痕，则硬度小于 5.5，否则硬度大于 5.5；

将矿物在普通硅酸盐玻璃（硬度为 5.5-6）上刻划，如不能将玻璃刻划出刻痕，在玻璃上留下矿物的粉末，则矿物硬度小于 5.5，如能将玻璃刻划出刻痕，则矿物硬度大于 5.5。

利用莫氏硬度计，通过矿物间的相互刻划，也可得出待鉴定矿物的大致硬度。利用该方法，须采用新鲜矿物，不能采用风化矿物。

莫氏硬度计

矿物	滑石	石膏	方解石	萤石	磷灰石
硬度	1	2	3	4	5
矿物	长石	石英	黄玉	刚玉	金刚石
硬度	6	7	8	9	10

（3）条痕：矿物粉末的颜色。

用矿物刻划普通硅酸盐玻璃或无釉瓷板，肉眼观察粉末的颜色。通过肉眼观察区分彩色条痕和无色（浅色）条痕。

应注意确定普通硅酸盐玻璃或无釉瓷板的粉末是否为矿物粉末，若矿物硬度较大，粉末可能为普通硅酸盐玻璃或无釉瓷板的粉末。故鉴别条痕后，应将粉末抹去，如普通硅酸盐玻璃或无釉瓷板上有刻痕，则应认定粉末不是矿物粉末，条痕鉴定结果无效。

（4）解理：矿物受外力后沿一定结晶平面破裂的固有特性。

通过肉眼或简单放大镜观察，区分有无解理，必要时确定解理组数。

1.2 矿物鉴定方法和步骤

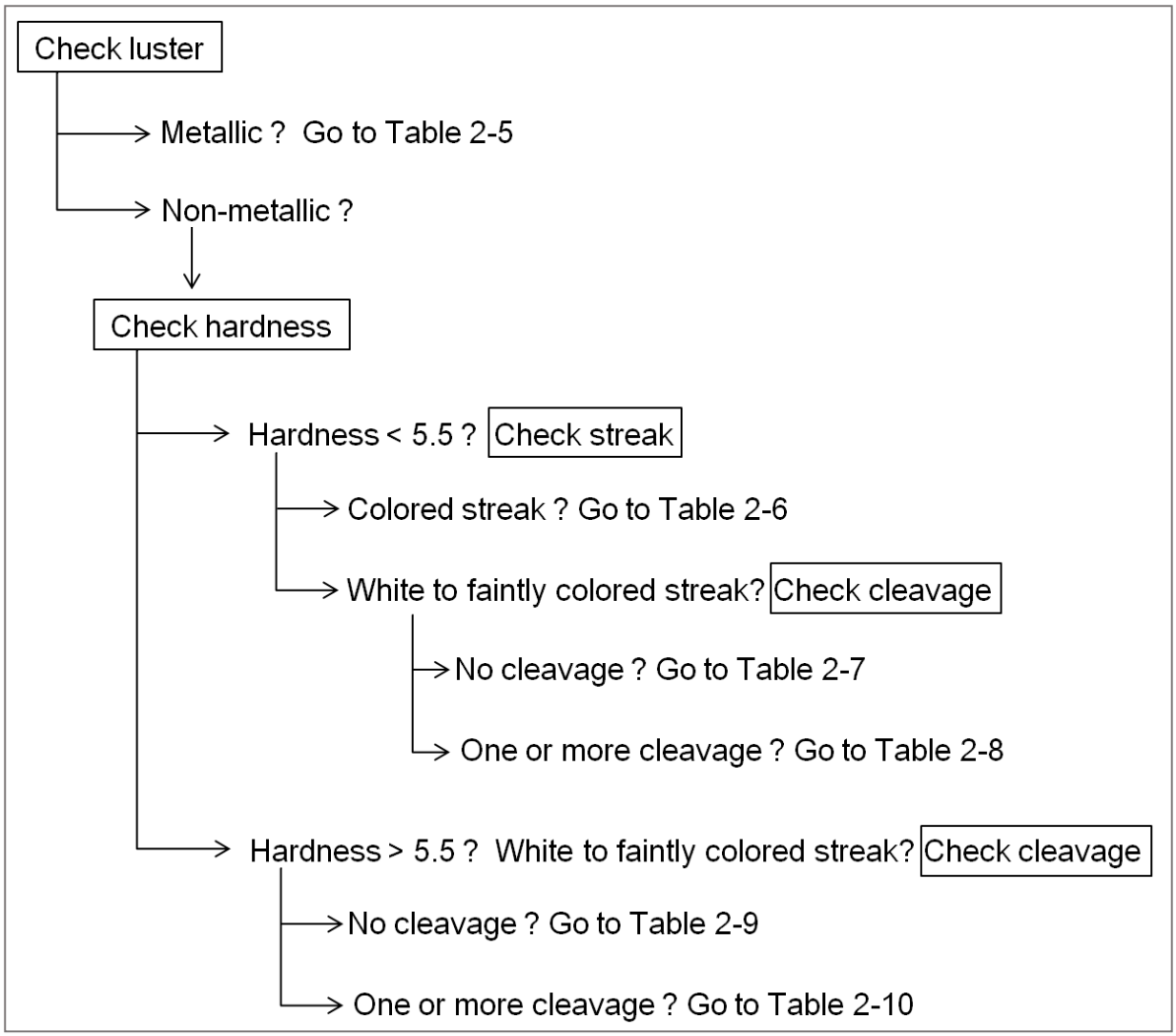
采用 N-C 方法进行矿物的鉴定，共分 4 步：

(1) 鉴定光泽，确定矿物的关泽是“金属光泽”或“非金属光泽”

(2) 鉴定硬度，确定矿物的硬度是“> 5.5”或 “< 5.5”；

(3) 鉴定条痕，确定矿物的条痕是“彩色”或“白色”；

(4) 鉴定解理，确定矿物 “无解理” 或“一组及多组解理”。



矿物鉴定 N-C 方法及程序

1.3 矿物鉴定表格

表 2-5 金属光泽矿物的鉴定

Hardness	Streak	Cleavage	Other properties	Mineral
Less than 5.5	Black, greenish black, brownish black, or gray	One (but not always visible)	Gray to black; H = 1; S.G. = 2.5; may have dull luster; greasy feel; soils paper and fingers, also in Table 2-6.	Graphite (C)
		Three (excellent cubic cleavage)	Silver gray, H = 2.5; S.G. =7.5; cubic crystals common.	Galena (PbS)
		None	Black to brownish black; H = 5.5; S.G. = 4.6; luster may be almost nonmetallic; brownish-black streak; weakly magnetic.	Chromite ($FeCr_2O_4$)
		None	Golden yellow or greenish yellow; H = 3.5~4; S.G. = 4.2; massive.	Chalcopyrite ($CuFeS_2$)
	Yellow-brown	None	Dark brown to black; H = 5~5.5; S.G. =4; massive or rounded forms with radiating, fiber-like layers.	Goethite FeO(OH)
	Copper-red	None	Copper to brown, may have green coating; H = 2.5~3; S.G. = 8.9; malleable, sectile.	Native copper (Cu)
Greater than 5.5	Greenish black to black	None	Brass-yellow; H = 6~6.5; S.G. = 5; massive or as crystals (cube, pyritohedron)	Pyrite (FeS_2)
	Black	None	Dark-gray to black; H = 6; S.G. = 5.2; may have dull luster if surface not fresh; attracted to magnet.	Magnetite (Fe_3O_4)
	Red-brown	None	Steel gray to dull red; H = 6; S.G. = 5; may be micaceous (tiny flakes) or massive, also in table 2-6.	Hematite (Fe_2O_3)

表 2-6 非金属光泽、硬度<5.5、深色条痕矿物的鉴定

Streak	Cleavage	Other properties	Mineral
Gray-black	One (but not always visible)	Gray to black; H = 1; S.G. = 2.5; may have dull luster; greasy feel; soils paper and fingers, also in Table 2-5.	Graphite (C)
Red-brown	None	Red to reddish brown; H = 1.5~5.5; S.G. = 5; dull luster; earthy or oolitic (containing spherical structures having diameters of 0.25 to 2 mm) masses, also in table 2-5.	Hematite (Fe_2O_3)
Yellow-brown	None	Yellow-brown, orange-brown to dark brown; H = 1.5 ~ 5.5; S.G. = 3.6 ~ 4.0; dull luster; earthy, may be a rustlike coating. Not a true mineral, limonite is a general name for several rustlike, hydrous iron oxides.	Limonite (Various hydrous iron oxides)
Yellow	None	Yellow; H = 1.5~2.5; S.G. = 2.1; resinous or vitreous luster.	Sulfur (S)
	Six - may be difficult to count	Light yellow, yellowish brown, black; H = 3.5 ~ 4; S.G. = 4.0; resinous or vitreous luster; faint odor of hydrogen sulfide (rotten eggs)	Sphalerite (ZnS)
Blue	None	Azure blue; H = 3.5 ~ 4; S.G. = 3.8; occurs as coating, masses, or tiny crystals, commonly with malachite, and commonly as an accessory mineral.	Azurite $Cu_3(CO_3)_2(OH)_2$
Green	None	Bright green; H = 3.5 ~ 4; S.G. = 4; occurs as coating, masses, or tiny crystals, commonly with azurite, and commonly as an accessory mineral.	Malachite $Cu_2CO_3(OH)_2$

表 2-7 非金属光泽、硬度<5.5、白色或浅色条痕、无解理矿物的鉴定

Cleavage	Other properties	Mineral
None or Indistinct	White, may have a brownish tint, H = 1 ~ 2.5; S.G. = 2.6; dull luster; greasy feel, earthy odor, powdery.	Kaolinte ($Al_2Si_2O_5(OH)_4$)
	White, gray, or apple green; pearly luster; H = 1 (or 2 in some impure varieties); S.G. = 2.7~2.8; greasy feel; may have light gray streak; also in Table 2-8.	Talc ($Mg_3Si_4O_{10}(OH)_2$)
	White, H = 2; S.G. = 2.3; satiny or silky luster and fibrous fracture indicates satin spar variety of gypsum. An opaque, white or gray, massive variety is alabaster; also in Table 2-8.	Gypsum ($CaSO_4\ 2H_2O$)
	Reddish brown to brown color; H = 1 ~ 6; S.G. = 2.0~2.5; dull to earthy luster; massive or with pea-shaped structures; may scratch glass with difficulty. Not a true mineral, but a rock consisting of several similar minerals. May have faint, red-brown streak, also in Table 2-9.	Bauxite (Mixture of hydrous aluminum oxides)
	Multicolored green, gray, black; H = 3 ~ 5; S.G. = 2.5~2.6; dull to greasy luster; slight greasy feel; massive to fibrous (asbestos).	Serpentine ($Mg_3Si_2O_5(OH)_4$)
	Light green to medium green, brown, yellow; vitreous luster; H = 5; S.G. = 3.2; six-sided crystals common; may show one poor cleavage.	Apatite $Ca_5(PO_4)_3(F,Cl,OH)$
	Buff, gray, white, pinkish; H = 3.5 ~ 4; S.G. = 2.8~2.9; small, rhombohedral crystals or massive; three cleavages, not at 90 °, may be indistinct in massive varieties; reacts slowly or not at all with dilute hydrochloric acid unless powdered. Also in Table 2-8.	Dolomite ($CaMg(CO_3)_2$)

表 2-8 非金属光泽、硬度<5.5、白色或浅色条痕、有解理矿物的鉴定

Cleavage	Other properties	Mineral
One	Light apple green, white, gray; pearly luster; H = 1; S.G. = 2.7~2.8; cleavage not evident in finely crystalline, massive varieties; may have light gray streak; greasy feel; also in Table 2-7.	Talc ($Mg_3Si_4O_{10}(OH)_2$)
	Green to blackish green; dull to pearly or vitreous; H = 2~2.5; S.G. = 2.6~3.3; may have faint green-yellow streak; cleavage flakes are flexible but not elastic; finely crystalline aggregates common.	Chlorite (Mg-Fe-Al silicate)
	Black to brownish black; vitreous luster; H = 2.5~3; S.G. = 2.8~3.2; may have faint brown-gray streak; individual crystals commonly are small and cleavage surface are wavy; perfect cleavage; transparent, flexible and elastic in thin sheets.	Biotite (Hydrous K-Mg-Fe-Al silicate)
	Colorless, silvery white, brownish silvery white; vitreous luster; H = 2 ~ 2.5; S.G. = 2.8~2.9; one perfect cleavage; transparent, flexible and elastic in thin sheets.	Muscovite (K-Al silicate)
	Clear, white, light gray; H = 2; S.G. = 2.3; vitreous to pearly luster; brittle sheets; one perfect cleavage, two poor cleavages indicate the selenite variety of gypsum; alabaster is massive, satin spar is fibrous, also in Table 2-7.	Gypsum ($CaSO_4 \cdot 2H_2O$)
Two	Black; H = 5~6; S.G. = 3~3.4; vitreous luster; two perfect cleavages meet at 124 °and 56 °; cleavage faces stepped rather than smooth; splintery appearance; may have greenish black to black streak. An amphibole also listed in Table 2-10.	Hornblende (silicate minerals)
	Black to dark green; H = 5~6; S.G. = 3.2~3.4; vitreous to dull luster; two imperfect cleavages meet at nearly 90 °. Tend to look "blocky", not splintery. A pyroxene. Another pyroxene, diopside, is similar but is light grayish green. Also in Table 2-10.	Augite (Ca-Mg-Fe silicate)
Three	Clear to gray and red; H = 2.5; S.G. = 2.2; three perfect cleavages meet at nearly 90 °(cleavage surfaces may dull and partially dissolve with prolonged exposure); salty tast.	Halite (NaCl)
	Clear, white, light gray; H = 2; S.G. = 2.3; vitreous to pearly luster; brittle sheets; one perfect cleavage, two poor cleavages may be evident in clear pieces. Also in Table 2-7.	Gypsum ($CaSO_4 \cdot 2H_2O$)
	Colorless or white; H = 3~3.5; S.G.= 4.5-heavy for a nonmetallic; tabular crystals, roselike array of crystals, or massive; three cleavages meet at 90 °.	Barite ($BaSO_4$)
	Clear, white, less commonly other colors; vitreous luster; H = 3; S.G. = 2.7; three perfect cleavages form rhombic cleavage fragments; double images seen through clear pieces; reacts strongly with dilute hydrochloric acid.	Calcite ($CaCO_3$)
	Buff, gray, white, pinkish; H = 3.5 ~ 4; S.G. = 2.8~2.9; small, rhombohedral crystals or massive; three cleavages, not at 90 °, may be indistinct in massive varieties; reacts slowly or not at all with dilute hydrochloric acid unless powdered. Also in Table 2-7.	Dolomite ($CaMg(CO_3)_2$)
Four	Purple, green, blue, yellow, clear; vitreous luster; H = 4; S.G. = 3.2; perfect cleavage up to four directions may yield octahedral cleavage fragments; may occur as cubic crystals.	Fluorite (CaF_2)

表 2-9 非金属光泽、硬度>5.5、无解理矿物的鉴定

Cleavage	Other properties	Mineral
None	Brown, pink, blue, gray; H = 9; S.G. = 4; Six-sided prismatic crystals: ruby (red) and sapphire (commonly blue) are gem varieties.	Corundum (Al_2O_3)
	Black, pink, blue, green, brown; vitreous luster; H = 7~7.5; S.G. = 3~3.3; slender crystals with triangular cross sections and striated sides.	Tourmaline (Complex hydrous silicate)
	Reddish brown, yellowish tan; vitreous luster; H= 6.5~7.5; S.G. = 3.6 ~ 4.3; twelve sided crystals common. Broken surfaces may resemble cleavage in some samples.	Garnet (Ca-Mg-Fe-Al silicate)
	Red-brown to brownish black; vitreous luster, or dull luster; H= 7~7.5; S.G. = 3.7; prismatic and X- or cross-shaped crystals.	Staurolite (Fe-Al silicate)
	Coarsely crystalline varieties: clear, milky, white, purple, smoky, pink; transparent to translucent; vitreous luster; H = 7; S.G. = 2.7; conchoidal fracture; usually massive; sometimes occurs as six sided crystals; microcrystalline varieties; chert (gray, dull luster), flint (black, dull luster), chalcedony (brown to gray, translucent, waxy luster), agate and anyx (yaricolored bands, vitreous luster).	Quartz (SiO_2)
	Colorless, white, or pale shades of yellow, green, red, or blue, may show play of colors; vitreous to resinous luster; H = 5~6; S.G. = 2~2.3; rounded forms common, but also massive; conchoidal fracture. A mineraloid.	Opal ($SiO_2 \cdot nH_2O$)
	Olive green to yellow green; vitreous to dull luster; H = 6.5~7 but difficult to test because of granular; S.G. = 3.3~4.4;	Olivine $(Mg, Fe)_2SiO_4$
	Reddish brown to brown color; H = 1~6; S.G. = 2.0~2.5; dull to earthy luster; massive or with pea-shaped structures; may scratch glass with difficulty. Not a true mineral, but a rock consisting of several similar minerals. Also in Table 2-7.	Bauxite (Mixture of hydrous aluminum oxides)

表 2-10 非金属光泽、硬度>5.5、有解理矿物的鉴定

Cleavage	Other properties	Mineral
One	Green, blue, cyan, yellow, colorless (pure beryl), red and others; H = 7.5~8; S.G. = 2.7~2.9; prismatic to tabular, radial, columnar crystals; granular to compact massive; vitreous luster; mostly transparent; one poor cleavage; an optical reflectance of chatoyancy; white streak; brittle; beryl of various colors is found most commonly in granitic pegmatites, or in mica schists and limestone; varieties of beryl are well-known gemstones including emerald (green beryl), aquamarine (blue or cyan bery), goshenite (colorless), golden beryl and red beryl.	Beryl $Be_3Al_2(SiO_3)_6$
	Colorless (pure topaz), blue, brown, orange, gray, yellow, green, pink and reddish pink; H = 8; S.G. = 3.3; the hardest silicate mineral; prismatic crystals with faces striated parallel to long dimension; columnar, compact, massive; vitreous luster; transparent; white streak; one perfect cleavage; topaz is commonly associated with siliceous igneous rocks of the granite and rhyolite type; topaz is wide used in jewellery as a cut gemstone due to its hardness and transparence;	Topaz $Al_2SiO_4(F,OH)_2$
Two	Black to dark green; H = 5~6; S.G. = 3.2~3.4; vitreous to dull luster; two imperfect cleavages meet at nearly 90°. Tend to look "blocky", not splintery. A pyroxene. Another pyroxene, diopside, is similar but is light grayish green. Also in Table 2-8.	Augite (Ca-Mg-Fe silicate)
	Black; H = 5~6; S.G. = 3~3.4; vitreous luster; two perfect cleavages meet at 124° and 56°; cleavage faces stepped rather than smooth; splintery appearance; may have greenish black to black streak. An amphibole also listed in Table 2-10.	Hornblende (silicates)
	Colorless, greenish, grayish yellow, white, pink; H = 6; S.G. = 2.5; columnar or tabular crystals; anhedral or euhedral; vitreous luster; transparent to translucent; white streak and uneven fractures; two perfect cleavages at 90°; low negative relief under microscope; orthoclase is commonly associated with quartz in granites; orthoclase commonly alters to sericite or clay;	Orthoclase ($KAlSi_3O_8$)
	White, gray, bluish white, reddish white, greenish white; H = 6~6.5; S.G.= 2.6~2.8; columnar or tabular crystals; vitreous luster; transparent to translucent; white streak; two perfect cleavages at 86°, or one perfect cleavage plus one good cleavage; plagioclase is a series of tectosilicate (framework silicate) minerals within the feldspar group, including anorthite (90-100 An%), bytownite (70-90 An%), labradorite (50-70 An%), andesine (30-50 An%), oligoclase (10-30 An%) and albite (0-10 An%); plagioclase is commonly coexisted with amphibole and augite in dark-colored igneous rocks, such as diorite and gabbro.	Plagioclase $NaAlSi_3O_8$-$CaAl_2Si_2O_8$
	Reddish brown, yellow, green, blue, gray, colorless; H = 7.5; S.G.= 4.6~4.7; tabular to prismatic crystals; vitreous to adamantine luster; white streak; transparent to opaque; conchoidal to uneven fracture; two poor cleavages; zircon is an accessory mineral in igneous and metamorphic rocks, with crystals of large sizes in mafic pegmatites and carbonatites; zircon contains trace amounts of uranium and thorium (from 10 ppm up to 1 wt%), and it can be dated using several modern analytical techniques.	Zircon ($ZrSiO_4$)
Three	Reddish brown, red, pale yellow, pale blue, violet, rarely grass-green; H = 6~6.5; S.G.= 4.23, increasing with Nb-Ta content; acicular to prismatic, elongated and striated crystals; adamantine luster; opaque; white or bright red to dark red streak; brittle; one perfect cleavage plus two good cleavages; strongly anisotropism; rutile is a common high-temperature and high-pressure accessory mineral in anorthosite, and granite pegmatite; rutile has one of the highest refractive indices at visible wavelengths of any known crystal and also exhibits a particularly large birefringence and high dispersion.	Rutile (TiO_2)

2. 岩浆岩鉴定实习

岩浆岩的鉴定命名:成分+结构+构造

（1）判断基本岩性：

1）看颜色，判断是酸性岩、中性岩、基性岩还是超基性岩；

2）看岩石结构，判断是深成岩、浅成岩还是喷出岩；

（2）岩石详细定名：

1）鉴别主要矿物及次要矿物；

2）鉴别岩石构造。

（3）为什么会有结构上的规律性？

岩浆自岩浆房向地表运移过程中，温度逐渐下降；

矿物在高温条件下结晶，速度慢，结晶程度好，粒度均匀；

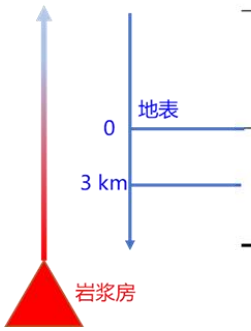
矿物在低温条件下结晶，速度快，结晶程度差，粒度不均；

喷出地表的岩浆，迅速冷凝，其中的组分来不及结晶，形成玻璃质。

（4）深入理解岩浆岩分类表，认识表中代表性岩石。

岩浆岩分类表

颜色					浅←——→深				
岩浆类型					酸性	中性		基性	超基性
SiO_2含量（%）					>65	65～52		52～45	<45
成因类型		产状	构造	主要矿物 / 次要矿物 / 结构					
				主要矿物	石英 正长石 斜长石	正长石 斜长石	角闪石 斜长石	斜长石 辉石	橄榄石 辉石
				次要矿物	云母 角闪石	角闪石 黑云母 辉石 石英<5%	辉石 黑云母 正长石、石英<5%	橄榄石 角闪石 黑云母	角闪石 斜长石 黑云母
喷出岩		岩钟 岩流	杏仁状 气孔状 流纹状 块状	非晶质（玻璃质）	火山玻璃：黑曜岩、浮岩等				少见
				隐晶质 斑状	流纹岩	粗面岩	安山岩	玄武岩	少见
侵入岩	浅成	岩床 岩墙	块状	斑状 全晶细粒	花岗斑岩	正长斑岩	闪长玢岩	辉绿岩	少见
	深成	岩株 岩基		结晶斑状 全晶中、粗粒	花岗岩	正长岩	闪长岩	辉长岩	橄榄岩 辉岩



3. 沉积岩鉴定实习

沉积岩的鉴定命名:结构+构造+成分

（1）判断基本岩性：

1）看结构，判断是碎屑岩、黏土岩还是化学及生物化学岩；

2）看构造，判断是泥岩还是页岩；

（2）岩石详细定名：

1）鉴别主要矿物构成，尤其是碎屑岩；

2）化学方法区分化学岩，如石灰岩和白云岩。

（3）深入理解沉积岩分类表，认识表中代表性岩石。

沉积岩分类表

<table>
<tr><th>分类</th><th>岩石名称</th><th colspan="2">结　构</th><th>构 造</th><th colspan="2">矿　物　成　分</th></tr>
<tr><td rowspan="5">碎
屑
岩</td><td>角砾岩</td><td rowspan="2">砾 状
结 构
(＞2 mm)</td><td>角砾状结构
(＞2 mm)</td><td rowspan="5">层
理
或
块
状</td><td rowspan="2">砾石成分为原岩碎屑成分</td><td rowspan="5">胶结物成分可为：硅质、钙质、铁质、泥质、碳质等</td></tr>
<tr><td>砾　岩</td><td>砾状结构
(＞2 mm)</td></tr>
<tr><td>粗砂岩</td><td rowspan="3">砂
状
结
构
(2～0.005 mm)</td><td>粗砂状结构
(2～0.5 mm)</td><td rowspan="3">砂粒成分：
① 石英砂岩：石英占 95%以上
② 长石砂岩：长石占 25%以上
③ 杂砂岩：含石英、长石及多量暗色矿物</td></tr>
<tr><td>中砂岩</td><td>中砂状结构
(0.5～0.25 mm)</td></tr>
<tr><td>细砂岩</td><td>细砂状结构
(0.25～0.075 mm)</td></tr>
<tr><td rowspan="2">黏
土
岩</td><td>页　岩</td><td colspan="2" rowspan="2">泥状结构
(＜0.005 mm)</td><td rowspan="2">页理
块状</td><td colspan="2" rowspan="2">颗粒成分为黏土矿物，并含其他硅质、钙质、铁质、碳质等成分</td></tr>
<tr><td>泥　岩</td></tr>
<tr><td rowspan="7">化
学
岩
及
生
物
化
学
岩</td><td>石灰岩</td><td colspan="2" rowspan="7">化学结构
及
生物化学结构</td><td rowspan="7">层
理
或
块
状
或
生
物
状</td><td colspan="2">方解石为主</td></tr>
<tr><td>白云岩</td><td colspan="2">白云石为主</td></tr>
<tr><td>泥灰岩</td><td colspan="2">方解石、黏土矿物</td></tr>
<tr><td>硅质岩</td><td colspan="2">燧石、蛋白石</td></tr>
<tr><td>石膏岩</td><td colspan="2">石膏</td></tr>
<tr><td>岩　盐</td><td colspan="2">NaCl、KCl 等</td></tr>
<tr><td>有机岩</td><td colspan="2">煤、油页岩等含碳、碳氢化合物的成分</td></tr>
</table>

4. 变质岩鉴定实习

变质岩的鉴定命名:构造+结构+成分

（1）判断基本岩性：

1）看岩石构造，判断岩石是否具有片理构造，如有则按片理构造类型鉴定岩石；

片理构造指岩石中矿物呈定向平行排列的构造。它是多数变质岩区别于岩浆岩和沉积岩的重要特征。根据变质程度由浅入深分为以下四种主要构造形式：

板状构造：泥质岩和砂质岩在定向压力作用下，产生一组平坦的破碎面，岩石易沿该面剥成薄板。

千枚状构造：矿物重结晶，岩石片理清楚，片里面上有许多定向排列的绢云母，呈明显的丝绢光泽。

片状构造：重结晶作用明显，片状、针状矿物沿片理面富集、平行排列。

片麻状构造：深色的片状及柱状矿物数量少，呈不连续的条带状，中间被浅色粒状矿物隔开。

2）看岩石结构，对于无片理构造的岩石，通过结构鉴定矿物变质的类型及变质程度。

（2）岩石详细定名：

鉴别主要矿物组成。

（3）深入理解变质岩分类表，认识表中代表性岩石。

变质岩分类表

变质作用	岩石名称	结构	构造		主要矿物成分
区域变质	板　岩	变余	片理构造	板　状	黏土矿物、云母、绿泥石、石英、长石等
	千枚岩	变余		千枚状	绢云母、石英、长石、绿泥石、方解石等
	片　岩	变晶		片　状	云母、角闪石、绿泥石、石墨、滑石等
	片麻岩	变晶		片麻状	石英、长石、云母、角闪石、辉石等
混合岩化	混合花岗岩	交代		片麻状	石英、长石
接触变质或区域变质	大理岩	变晶	块状构造		方解石、白云石
	石英岩	变晶			石英
	矽卡岩	变晶			石榴子石、辉石、方解石
交代变质	云英岩	变晶			白云母、石英
	蛇纹岩	隐晶			蛇纹石
动力变质	断层角砾岩	压碎			岩石、矿物碎屑
	糜棱岩	糜棱			石英、长石、绿泥石、绢云母

5. 三大岩类的鉴别

可通过以下特征区别鉴定三大岩类，但并不绝对，应根据具体岩样具体分析，必要时应采用显微镜薄片鉴定等技术。

岩浆岩：矿物结晶程度高、光泽好；颗粒棱角状和次棱角状、无胶结而可能有熔结。

沉积岩：有层理；有生物结构；颗粒有磨圆、经过了分选；颗粒光泽感不强。

变质岩：矿物有重新结晶、光泽好；颗粒的定向排列；岩体碎裂及糜棱化。

岩矿鉴定表格

矿物鉴定表格

姓名 _____班级_____ 学号

标本号	第一步 鉴定结果	第二步 鉴定结果	第三步 鉴定结果	第四步 鉴定结果	适用表格 序号	其他特征	鉴定结论

岩浆岩鉴定表格

姓名 _____班级_____ 学号_____

标本号	颜色	按 SiO_2 判定的岩石类型	结构特征	按结构判定的类型	构造特征	矿物组成	鉴定结论

沉积岩鉴定表格

姓名 _____班级_____ 学号_____

标本号	结构特征	结构划分的岩石类型（打 √）			构造特征	矿物组成	其他特征，如颜色等	鉴定结论
		碎屑岩	泥质岩	化学及生物化学岩				

变质岩鉴定表格

姓名 _____班级_____ 学号_____

岩石代号	岩石构造	岩石结构	岩石矿物组成	其他性质	鉴定结论

三大岩类辨识

姓名 _____班级_____ 学号_____

标本号	结构特征	组成岩石的颗粒特征					是否有生物结构	鉴定结论
		是否有磨圆	是否有分选	是否重新结晶	是否定向排列	是否糜棱化		

6 岩石耐磨性测试实习

岩石耐磨性作为岩石的基本力学指标之一，目前在国内外学术界尚缺对该领域的系统化研究，但是随着岩石学科科研工作的不断进展，人们生产实践的需要，尤其是集掘进、排渣装运、衬砌施工等工序为一体的大型隧道施工设备（隧道掘进机）在隧道工程中的使用，使得人们开始重新认识到开展岩石耐磨性研究的必要性和重要性。岩石硬度测定方法繁多，就目前而言，以压入试验、动力或回弹试验、划痕实验为主，其中操作最方便快捷的当属划痕实验。试验原理如下：

岩石的耐磨性是破岩工具在岩石的正摩擦力作用下，达到工具的极限屈服点和极限强度时，摩擦力在工具和岩石接触面上产生了应力，从而引起工具表面的剪切破碎的现象，而岩石的耐磨性代表了造成破岩工具破碎的能力。国际岩石力学学会（ISRM）提出了 Cerchar 研磨性指标（CAI）测试作为岩石研磨性的实验方法，这是应用最广泛的测试手段。本实习指导书运用了此方法原理，钢针材料采用洛氏硬度为 55 的合金 45 号钢，整个实验装置在实验过程中保持平稳不晃动，使钢针在 7.0kg 的圆柱形荷载的作用下垂直于岩芯表面水平划过 10mm 并留下一定深度的凹槽，然后采用电子显微镜测量钢针针尖的磨损量，以其磨损量的大小来表征实验岩样的耐磨性强弱。

试验步骤如下：

1、启动电源，通过电机控制面板上操作调节岩样夹持平台位置以利于安装试件。

2、安装试件，保持测试面水平、加持面固定，将岩样放入虎钳中，转动虎钳手柄夹紧岩样至不晃动即可。

3、安装测试钢针，将钢针放入夹具内拧紧夹具，在钢针夹持固定后缓缓旋下主机荷重，使针尖垂直压在测试面表面，并注意在安装钢针并放置到试样表面上时一定要轻，避免放针过猛导致针尖被撞磨而造成人为误差。

4、按下测试开始按钮，自动开始划刻。

5、试验完成后拿下试样和钢针，完成单块岩石试样第一次划痕。

6、数值测量，采用“三分旋转测量法”进行测读，即每根钢针测读 3 次，每次测读角度应轴向旋转 120°，以 3 次测量的平均值作为该钢针针尖磨损后直径测量值，以符号 dw 表示，记录单位为μm，测量精度为 0.01 μm。

岩石耐磨性指数 *CAI* 按下式计算：

$$CAI' = (\frac{1}{n'}\sum_{i=1}^{n'} A_b) \cdot 10^{-2} \quad \text{（式 1）}$$

$$A_b = d_w - d_i \quad \text{（式 2）}$$

式中 CAI'—单个岩石试样自然表面耐磨性指数，计算结果按照四舍五入原则保留 2 位小数，单位 mm；

A_b — 针尖磨损值，即试验前后针尖直径变化值，单位μm;

d_i—钢针初始值，即试验前针尖直径测量值，单位μm;

d_w—钢针试验值，即试验后针尖直径测量值，单位μm;

n' — 单个岩石试样表面划痕数量。

根据岩石耐磨性指数标准值 CAI 确定岩石耐磨强度等级，等级划分详见下表。

岩石耐磨性等级划分表

岩石耐磨性等级	极低耐磨性	低耐磨性	中耐磨性	强耐磨性	极强耐磨性
岩石耐磨性指数 CAI（0.1mm）	<1	1－2	2－3	3－4	>4

注：当岩石耐磨性指数标准值 CAI 为界限值时，按照不利原则判定岩石耐磨强度等级。

耐磨性记录表格

试样编号	初始值 d_i（μm）	磨损后三分旋转（120°）测量值（μm）			试验值 d_w（μm）	针尖磨蚀值 A_b（μm）	耐磨性等级
	备注：$A_b = d_w - d_i$ A_b— 针尖磨损值，即试验前后针尖直径变化值，单位μm； d_i— 钢针初始值，即试验前针尖直径测量值，单位μm； d_w— 钢针试验值，即试验后针尖直径测量值，单位μm；						